

From Java to the Cloud: Integrating AI and Jetstream2 into Introductory Programming

How do we prepare beginning computer science students for a world where artificial intelligence and cloud computing are becoming essential parts of the computing profession? This question motivated the redesign of my introductory programming course. Traditionally, students learned programming concepts and completed assignments using local computers and installed software. While this approach provided a strong programming foundation, it offered limited opportunities for students to experience cloud computing, AI-assisted programming, and authentic research computing environments.

The redesigned course introduces students to these technologies progressively. Students begin by developing foundational Java programming skills before learning about responsible AI use and AI-assisted programming. They then transition to Jetstream2, where they gain experience working in a cloud-based computing environment. Later in the course, students use computational resources for programming, data analysis, visualization, and introductory AI projects.

A major goal of the redesign is to make advanced computing more accessible to beginning programmers. Instead of treating cloud computing and AI as advanced topics that students encounter later in their academic careers, the course provides an early introduction through hands-on activities. Students learn not only how to use AI to generate or explain code but also how to evaluate AI-generated solutions for accuracy, security, bias, and ethical considerations.

The project also emphasizes professional practices such as documentation, testing, debugging, reproducibility, and reflection. Students are assessed not simply on whether their programs work, but on whether they can explain their solutions, evaluate AI-generated code, document their processes, and demonstrate responsible use of computing resources.

Ultimately, the goal is to create a pathway from **programming fundamentals** → **AI literacy** → **cloud computing** → **Jetstream2** → **data and AI projects** → **career readiness**. This approach has the potential to expand beyond one course and provide a scalable model for integrating Science Gateway resources and advanced computing into additional computer science and STEM courses.